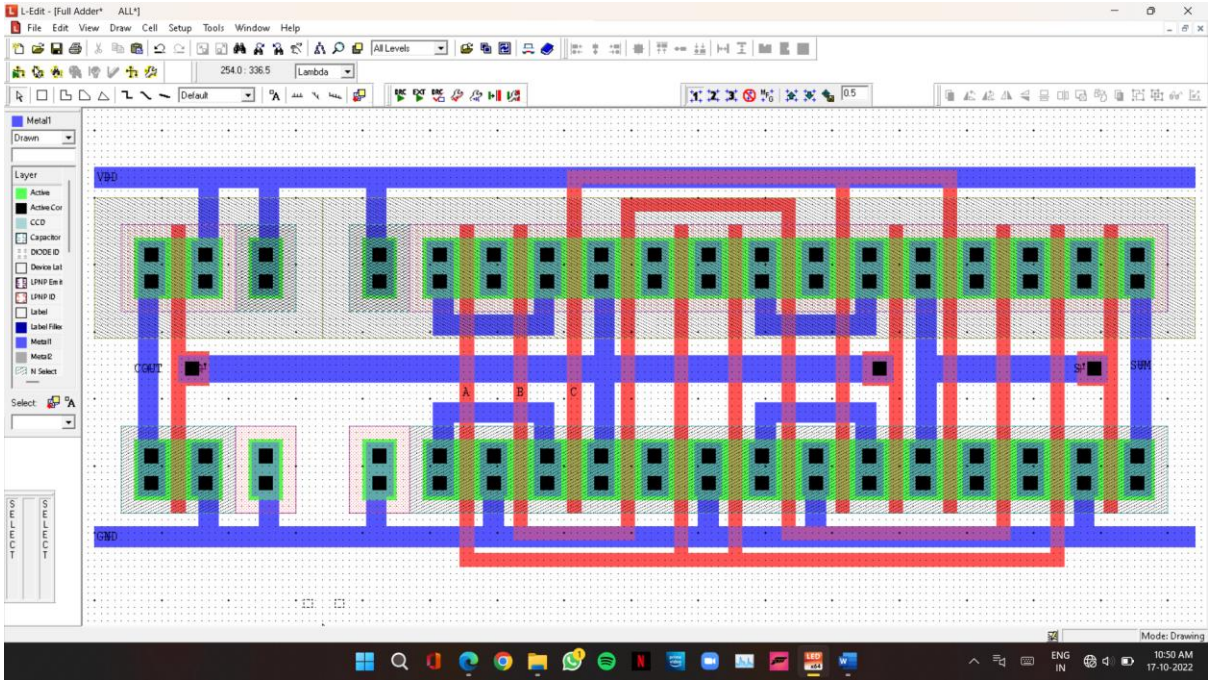
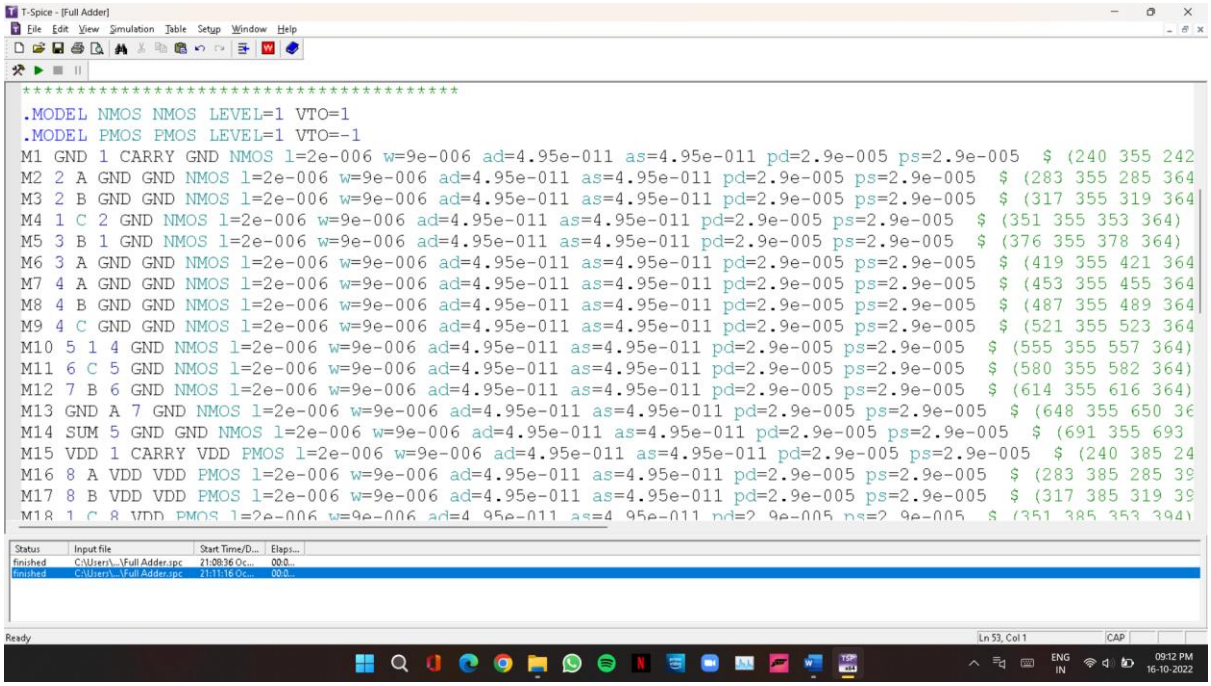
LAB – 12

28T Full Adder

K.N Sujay 121910402035

Outputs:





T-Spice - [Full Adder]

File Edit View Simulation Table Setup Window Help

```
M18 1 C 8 VDD PMOS l=2e-006 w=9e-006 ad=4.95e-011 as=4.95e-011 pd=2.9e-005 ps=2.9e-005  $ (351 385 353 394)
M19 9 B 1 VDD PMOS l=2e-006 w=9e-006 ad=4.95e-011 as=4.95e-011 pd=2.9e-005 ps=2.9e-005  $ (376 385 378 394)
M20 9 A VDD VDD PMOS l=2e-006 w=9e-006 ad=4.95e-011 as=4.95e-011 pd=2.9e-005 ps=2.9e-005  $ (419 385 421 39
M21 10 A VDD VDD PMOS l=2e-006 w=9e-006 ad=4.95e-011 as=4.95e-011 pd=2.9e-005 ps=2.9e-005  $ (453 385 455 3
M22 10 B VDD VDD PMOS l=2e-006 w=9e-006 ad=4.95e-011 as=4.95e-011 pd=2.9e-005 ps=2.9e-005  $ (487 385 489 3
M23 10 C VDD VDD PMOS l=2e-006 w=9e-006 ad=4.95e-011 as=4.95e-011 pd=2.9e-005 ps=2.9e-005  $ (521 385 523 3
M24 5 1 10 VDD PMOS l=2e-006 w=9e-006 ad=4.95e-011 as=4.95e-011 pd=2.9e-005 ps=2.9e-005  $ (555 385 557 394
M25 11 C 5 VDD PMOS l=2e-006 w=9e-006 ad=4.95e-011 as=4.95e-011 pd=2.9e-005 ps=2.9e-005  $ (580 385 582 394
M26 12 B 11 VDD PMOS l=2e-006 w=9e-006 ad=4.95e-011 as=4.95e-011 pd=2.9e-005 ps=2.9e-005  $ (614 385 616 39
M27 VDD A 12 VDD PMOS l=2e-006 w=9e-006 ad=4.95e-011 as=4.95e-011 pd=2.9e-005 ps=2.9e-005  $ (648 385 650 3
M28 SUM 5 VDD VDD PMOS l=2e-006 w=9e-006 ad=4.95e-011 as=4.95e-011 pd=2.9e-005 ps=2.9e-005  $ (691 385 693
VDD VDD GND DC 5
VG A GND DC 5
VG1 B GND DC 5
VG2 C GND DC 5
.DC VG 0 5 0.01
.PRINT V(SUM) V(CARRY) I(M1) I(M2) I(M3) I(M4) I(M5) I(M6) I(M7) I(M8) I(M9) I(M10) I(M11) I(M12) I(M13) I(
*VG A GND PULSE (0 5 0 1N 1N 10N 20N)
*VG1 B GND PULSE (0 5 0 1N 1N 20N 40N)
*VG2 C GND PULSE (0 5 0 1N 1N 40N 80N)
* TRAN 1N 400N
```

Status	Input file	Start Time/D...	Elaps...
finished	C:\Users\...\Full Adder.spc	21:08:36 Oc...	00:0...
finished	C:\Users\...\Full Adder.spc	21:11:16 Oc...	00:0...

Ready Ln 53, Col 1 CAP

ENG IN 09:13 PM 16-10-2022

T-Spice - [Full Adder]

File Edit View Simulation Table Setup Window Help

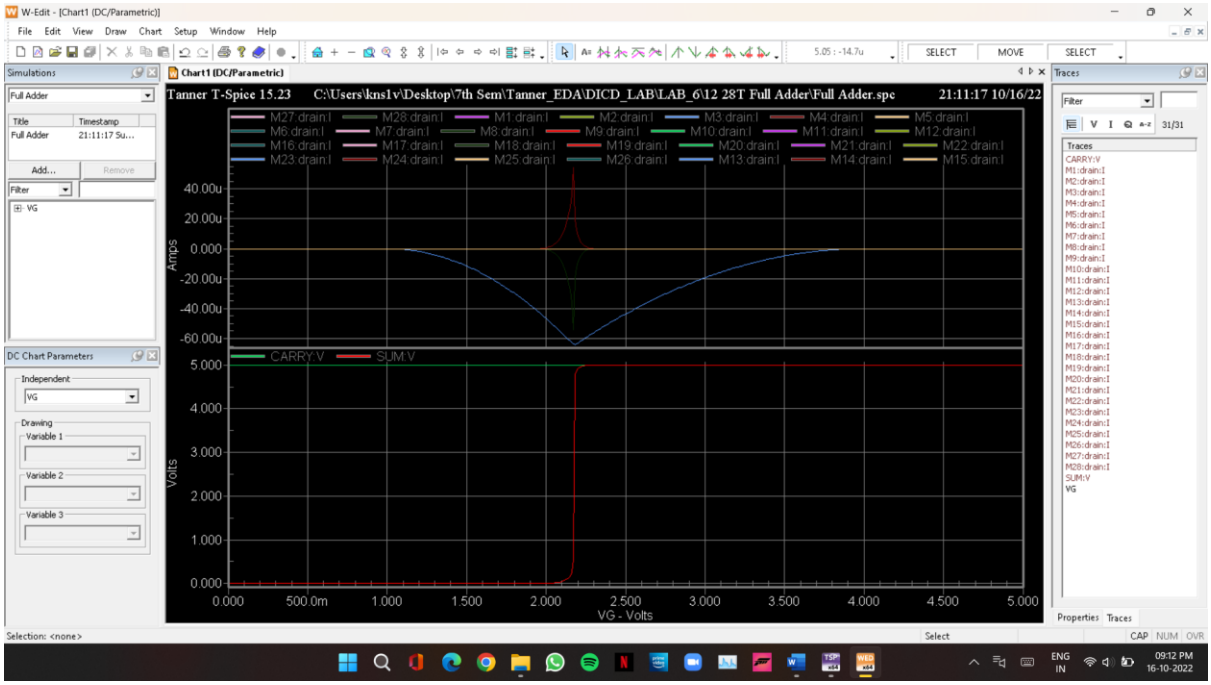
```
M26 12 B 11 VDD PMOS l=2e-006 w=9e-006 ad=4.95e-011 as=4.95e-011 pd=2.9e-005 ps=2.9e-005  $ (614 385 616 39
M27 VDD A 12 VDD PMOS l=2e-006 w=9e-006 ad=4.95e-011 as=4.95e-011 pd=2.9e-005 ps=2.9e-005  $ (648 385 650 3
M28 SUM 5 VDD VDD PMOS l=2e-006 w=9e-006 ad=4.95e-011 as=4.95e-011 pd=2.9e-005 ps=2.9e-005  $ (691 385 693
VDD VDD GND DC 5
VG A GND DC 5
VG1 B GND DC 5
VG2 C GND DC 5
.DC VG 0 5 0.01
.PRINT V(SUM) V(CARRY) I(M1) I(M2) I(M3) I(M4) I(M5) I(M6) I(M7) I(M8) I(M9) I(M10) I(M11) I(M12) I(M13) I(
*VG A GND PULSE (0 5 0 1N 1N 10N 20N)
*VG1 B GND PULSE (0 5 0 1N 1N 20N 40N)
*VG2 C GND PULSE (0 5 0 1N 1N 40N 80N)
*.TRAN 1N 400N
*.PRINT V(SUM) V(CARRY) V(A) V(B)  V(C)
.END
* Top level device count
* M(NMOS)       14
* M(PMOS)       14
* Number of devices: 28
* Number of nodes:   19
```

Status	Input file	Start Time/D...	Elaps...
finished	C:\Users\...\Full Adder.spc	21:08:36 Oc...	00:0...
finished	C:\Users\...\Full Adder.spc	21:11:16 Oc...	00:0...

Ready Ln 53, Col 1 CAP

ENG IN 09:13 PM 16-10-2022

W-Edit - [Chart1 (Transient)]
Chart1 (Transient)
Tanner T-Spice 15.23 C:\Users\kns1v\Desktop\7th Sem\Tanner_EDA\DICD_LAB\LAB_6\12 28T Full Adder\Full Adder.spc 21:08:36 10/16/22
SUM:V
CARRY:V
A:V
B:V
C:V
Volts
Seconds
0.000 50.00n 100.0n 150.0n 200.0n 250.0n 300.0n 350.0n 400.0n


W-Edit - [Chart1 (DC/Parametric)]
Chart1 (DC/Parametric)
Tanner T-Spice 15.23 C:\Users\kns1v\Desktop\7th Sem\Tanner_EDA\DICD_LAB\LAB_6\12 28T Full Adder\Full Adder.spc 21:11:17 10/16/22
Amps
CARRY:V
SUM:V
Volts
VG - Volts
0.000 500.0m 1.000 1.500 2.000 2.500 3.000 3.500 4.000 4.500 5.000